



How can India expand AYUSH education while ensuring credible training and effective quality regulation?

Capacity Gap • Clinical Exposure • Faculty Shortfalls • Regulatory Oversight • Incentive Misalignment

GS Paper II
Health
Government Policies

KNOWLEDGE POINTS

- **AYUSH Education System:** AYUSH medical education covers traditional and alternative systems such as Ayurveda and homoeopathy, requiring specialised colleges, teaching hospitals, qualified faculty, clinical exposure and appropriate regulatory oversight.
- **Private-Sector Expansion:** Private institutions increasingly drive AYUSH education growth, creating additional capacity but also raising concerns that financial incentives may encourage higher student intake without proportionate investment in faculty and infrastructure.
- **Quality Capacity Gap:** Rapidly increasing seats can strain educational quality when teaching staff, laboratories, hospital beds and clinical services do not expand proportionately, weakening students' practical exposure and institutional capacity.
- **Clinical Training Quality:** Medical education depends not merely on classroom teaching but meaningful clinical exposure; inadequate hospital facilities, patient attendance or bed occupancy can restrict students' opportunities to develop practical competence.
- **Quality-Control Reform:** Sustainable expansion requires stronger inspections, credible faculty verification, enforcement of admission limits and infrastructure standards, alongside regulatory mechanisms that align institutional incentives with educational quality.

QUICK FACTS

- In 2024, non-government institutions accounted for 86% of Ayurveda colleges and 85% of homoeopathy colleges, demonstrating private-sector dominance in AYUSH education.
- Between 2021 and 2024, permitted AYUSH seats increased by 43%, while total admission capacity rose by 25%, indicating rapid expansion of medical education.
- During 2021–2024, the Centre's AYURGYAN allocation for AYUSH education, training, research, innovation and capacity-building increased nearly sixfold alongside expanding educational capacity.
- A 2020 Journal of Ayurveda and Integrative Medicine article reported that many institutions had teaching-staff shortfalls exceeding 50% against the standards then applicable.

KEY TAKEAWAY

Expanding AYUSH education sustainably requires growth in seats to be matched by qualified faculty, clinical exposure, adequate infrastructure and rigorous regulatory oversight.

MAINS PERSPECTIVE

Examine the challenges posed by private-sector-led expansion of AYUSH medical education in India. Suggest measures to ensure quality and accountability?

India's AYUSH education system is expanding rapidly, with private institutions providing much of its educational capacity. While private participation improves access and availability of seats, expansion without corresponding improvements in teaching and clinical infrastructure can undermine educational quality and accountability.

Private-sector-led expansion creates several challenges. Admission-linked revenue may encourage institutions to increase student intake without proportionately strengthening faculty, laboratories, hospital beds and clinical facilities. Inadequate teaching staff and limited patient exposure can weaken practical training and professional competence. Regulatory deficiencies concerning inspections, faculty strength and admission requirements further complicate quality assurance. Addressing these concerns requires rigorous inspections, transparent institutional grading and credible verification of faculty and infrastructure. Regulators should enforce admission limits and link permitted seats to demonstrated teaching and clinical capacity. Public disclosure of inspection findings can strengthen accountability, while regulatory standards should ensure that expansion remains consistent with educational quality and patient welfare.

Private participation can substantially expand AYUSH education, but numerical growth cannot substitute for institutional capability. Aligning commercial incentives with rigorous academic, clinical and regulatory standards is essential for producing competent practitioners and maintaining public confidence.

DAILY MCQs

- With reference to AYUSH medical education in India, consider the following statements:
Non-government institutions accounted for more than four-fifths of Ayurveda colleges in 2024.
Non-government institutions accounted for more than four-fifths of homoeopathy colleges in 2024.
Permitted AYUSH seats declined between 2021 and 2024.
Which of the statements given above are correct?
A. 1 only
B. 1 and 2 only
C. 2 and 3 only
D. 1, 2 and 3
- Consider the following pairs:
Qualified faculty — Teaching capacity
Teaching hospitals — Clinical exposure
Laboratories — Practical training
Higher student intake — Automatic improvement in educational quality
How many of the above pairs are correctly matched?
A. Only one
B. Only two
C. Only three
D. All four
- In the context of private-sector-led expansion of professional education, "incentive misalignment" would most appropriately refer to:
A. Increasing institutional expenditure whenever student intake rises
B. Financial incentives encouraging greater admissions without proportionate investment in educational capacity
C. Regulators reducing admission capacity despite complete compliance
D. Students preferring private institutions over public institutions

Answer Key • 1-B • 2-C • 3-B



How can India protect recovering vulture populations from emerging threats posed by expanding power infrastructure?

Ecological Scavengers • Diclofenac Threat • Electrocutation Risk • Bird-Safe Infrastructure • Preventive Planning

GS Paper III

Environment
Biodiversity
Conservation and
Infrastructure

KNOWLEDGE POINTS

- **Ecological Scavengers:** Vultures rapidly consume animal carcasses, limiting their persistence in the environment and thereby helping regulate scavenger populations, disease transmission risks and broader ecosystem sanitation.
- **Electrocutation Mechanism:** Vultures can be electrocuted when their large wingspans simultaneously bridge energised conductors or connect energised and grounded components, allowing dangerous electrical current to pass through them.
- **Behavioural Vulnerability:** Vultures frequently perch on elevated structures and congregate around predictable food sources, making electrical installations particularly hazardous when carcasses or food waste attract birds nearby.
- **Infrastructure Expansion Risk:** Expanding electricity infrastructure can create new wildlife hazards when transmission and distribution systems are designed without considering bird behaviour, body dimensions and ecologically sensitive locations.
- **Bird-Safe Infrastructure:** Separating feeding sites from electrical infrastructure, insulating conductors, increasing component spacing and installing safe perches can reduce opportunities for vultures and other birds to complete electrical circuits.

QUICK FACTS

- India's vulture population declined by 99.5% by 2007 from around four crore in the 1980s, severely affecting several major vulture species.
- Diclofenac, a veterinary painkiller that causes kidney damage in vultures feeding on treated cattle carcasses, was identified as the principal cause of the historic population crash.
- India banned veterinary use of diclofenac, while additional bans introduced in 2023 targeted aceclofenac and ketoprofen to reduce pharmaceutical threats to vultures.
- An official survey reported in 2025 found vultures nesting at only 50% of their historic nesting sites, highlighting the continuing fragility of their recovery.

KEY TAKEAWAY

Infrastructure development and biodiversity conservation must advance together by integrating wildlife behaviour, ecological vulnerability and preventive safeguards into power-system planning.

MAINS PERSPECTIVE

"Infrastructure development must incorporate biodiversity safeguards at the planning and design stages." Discuss in the context of electrocution risks faced by vultures in India?

India's infrastructure expansion is essential for development and universal electricity access, but poorly designed power systems can impose ecological costs. Vulture electrocution demonstrates why biodiversity safeguards must be integrated into infrastructure planning rather than introduced after wildlife losses.

Vultures are vulnerable because their large wingspans can bridge conductors or energised and grounded components, causing electrocution. Their tendency to perch on elevated structures and congregate near predictable food sources further increases exposure. This is especially concerning because Indian vulture populations remain fragile after their historic collapse caused by veterinary diclofenac. Persistent mortality can therefore undermine recovery. Biodiversity safeguards should begin with ecological risk assessments identifying nesting, feeding and congregation sites before power infrastructure is located. Carcass dumps and feeding sites should be separated from dangerous electrical installations. In high-risk areas, conductors can be insulated, spacing between electrical components increased, and safe perches installed. Wildlife mortality should be monitored. Cooperation among power utilities, conservation agencies and local authorities can institutionalise preventive measures.

Development and conservation need not be competing objectives. Incorporating species behaviour and ecological vulnerability into infrastructure location, design and operation can prevent avoidable mortality while allowing electricity networks to expand sustainably and making biodiversity protection integral to responsible infrastructure governance.

DAILY MCQs

- **With reference to vultures in India, consider the following statements:**
India's vulture population declined by about 99.5% by 2007 from its level in the 1980s.
Veterinary diclofenac was a major cause of the population collapse.
Vulture electrocution can occur when a bird bridges two conductors simultaneously.
Which of the statements given above are correct?
A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3
- **Which of the following measures can reduce electrocution risks to vultures?**
Separating feeding sites from electrical infrastructure
Insulating dangerous conductors
Increasing spacing between energised and grounded components
Installing safe perches
Select the correct answer using the code below:
A. 1 and 2 only
B. 1, 2 and 3 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4
- **The vulture-electrocution problem best illustrates which principle of environmental governance?**
A. Environmental protection should begin only after measurable population decline occurs.
B. Infrastructure planning should incorporate species-specific ecological and behavioural risks before significant damage occurs.
C. Wildlife conservation requires stopping infrastructure development in biodiversity-rich regions.
D. Chemical pollution is necessarily a greater biodiversity threat than physical infrastructure.

Answer Key • 1-D • 2-D • 3-B